

Quiz day 11

1. Write the cyclic monomial that corresponds to the transposition $(12)(34)(5)(6789)$.
2. You know that the cycle index of the group G that acts on the colourings X is given by

$$CI(z_1, z_2, z_3, z_4) = \frac{1}{24}(z_1^6 + 6z_2z_4 + 8z_3^2 + 9z_1^2z_2^2)$$

- a) How many different colourings in 1, 2 or 3 colours are possible?
- b) How many different colourings in exactly 3 colours are possible?

You don't need to finalize all arithmetic calculations in the quiz, just write the answer as an expression with numbers.